

Problem

A circuit consisting of a coil with inductance 10 mH and resistance 20 Ω is connected in series with a capacitor and a generator with an rms voltage of 120 V. Find:

(a) the value of the capacitance that will cause the circuit to be in resonance at 15 kHz

(b) the current through the coil at resonance

(c) the Q of the circuit

Step-by-step solution

Step 1 of 6

(a)

Consider the circuit to be in resonance at frequency, $f_0 = 15 \text{ kHz}$.

Write the expression for resonant frequency.

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

Substitute 10 mH for L and 15 kHz for f_0 in the above expression.

$$15 \times 10^{-3} = \frac{1}{2\pi\sqrt{(10 \times 10^{-3})C}}$$

Rewrite the equation.

$$\left(2\pi\sqrt{(10 \times 10^{-3})C}\right) = \left(\frac{1}{15 \times 10^3}\right)$$

Square on both sides

$$(2\pi)^2 \left(\sqrt{(10 \times 10^{-3})C}\right)^2 = \left(\frac{1}{15 \times 10^3}\right)^2$$

$$(2\pi)^2 \left((10 \times 10^{-3})C\right) = \left(\frac{1}{15 \times 10^3}\right)^2$$

$$C = \frac{\left(\frac{1}{15 \times 10^3}\right)^2}{(2\pi)^2 (10 \times 10^{-3})}$$

$$\cong 11.257 \text{ nF}$$

Therefore at resonance condition the capacitor C is 11.257 nF.

Step 2 of 6

(b)

Consider the series resonant circuit:

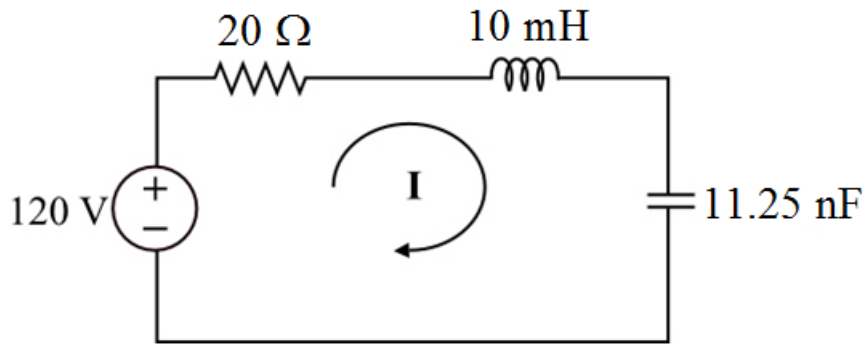


Figure 1

Step 3 of 6

First, convert the circuit to the frequency domain.

$$V_s = 120 \text{ V}$$

Determine the value of inductive reactance.

$$\begin{aligned} X_L &= j\omega L \\ &= j(2\pi f_0)L \\ &= j(2\pi(15 \times 10^3))(10 \times 10^{-3}) \\ &\cong j942.5 \Omega \end{aligned}$$

Determine the value of capacitive reactance.

$$\begin{aligned} X_C &= \frac{1}{j\omega C} \\ &= \frac{1}{j(2\pi f_0)C} \\ &= \frac{1}{j(2\pi(15 \times 10^3))(11.257 \times 10^{-9})} \\ &\cong -j942.5 \Omega \end{aligned}$$

Step 4 of 6

Thus, the frequency domain equivalent circuit is as shown below.

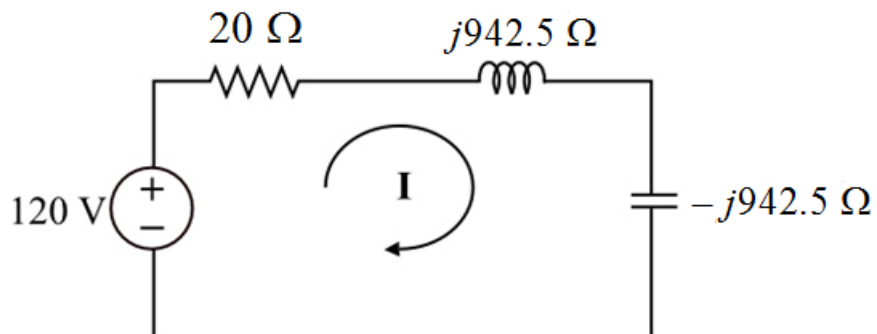


Figure 2

Step 5 of 6

Assume total impedance of the series resonant circuit is Z_1 .

$$\begin{aligned} Z_1 &= 20 + j942.5 - j942.5 \\ &= 20 \, \Omega \end{aligned}$$

Calculate the current through the coil at resonance using ohm's law.

$$\begin{aligned} I &= \frac{V_s}{Z_1} \\ &= \frac{120}{20} \\ &= 6 \, \text{A} \end{aligned}$$

Therefore, the current through the coil I at resonance is $\boxed{6 \, \text{A}}$.

Step 6 of 6

(c)

Write the expression for quality factor of the series resonant circuit.

$$\begin{aligned} Q &= \frac{1}{\omega_0 CR} \quad \text{or} \\ &= \frac{\omega_0 L}{R} \end{aligned}$$

Substitute 10 mH for L , $2\pi(15 \times 10^3)$ for ω_0 and $20 \, \Omega$ for R in the above expression.

$$\begin{aligned} Q &= \frac{\omega_0 L}{R} \\ &= \frac{(2\pi(15 \times 10^3))(10 \times 10^{-3})}{20} \\ &= 47.12 \end{aligned}$$

Therefore, the quality factor Q is $\boxed{47.12}$.